DTU

Model-based Machine Learning

Regression models

Outline

Case study: Modeling taxi demand in NYC

Linear regression

Poisson regression

Heteroscedastic regression

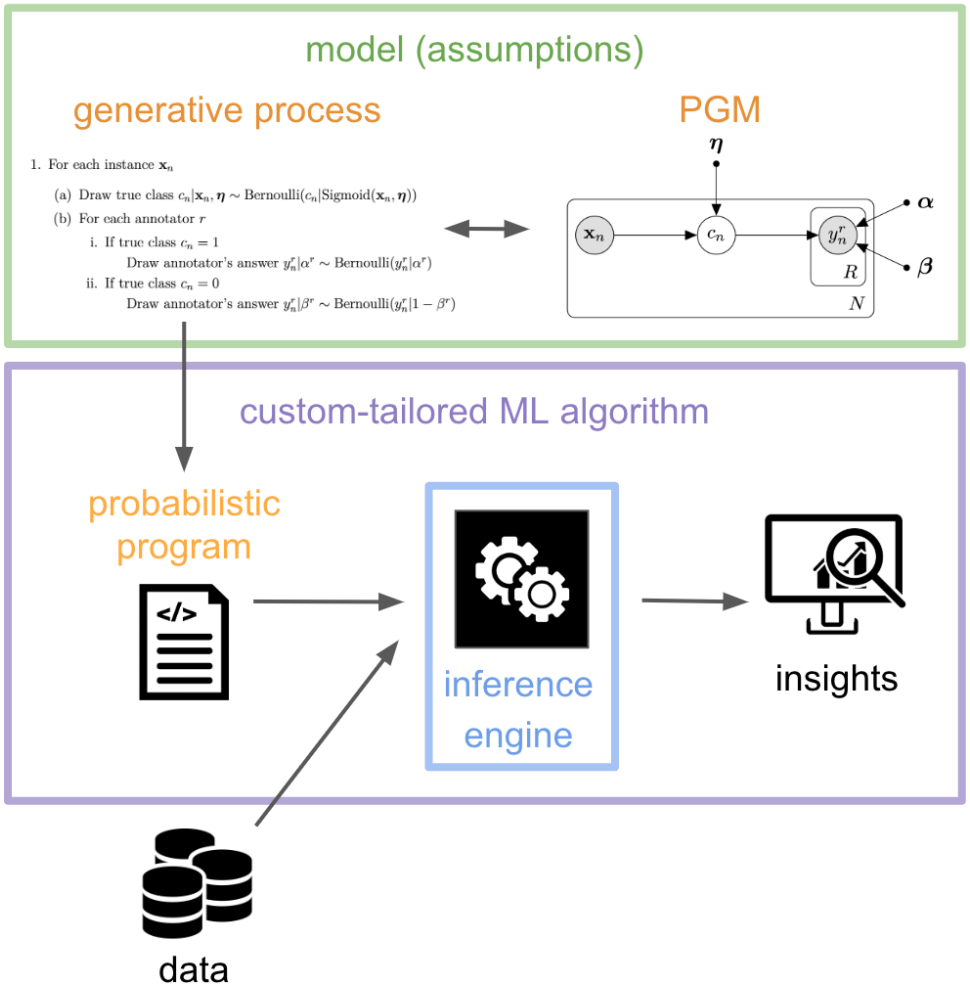
Non-linear models

Road to MBML: where are we?

Foundations of PGMs and probabilistic programming
Probabilistic modelling in various contexts - your modelling toolbox
Bayesian inference (exact and approximate)

Week	Topic
1	Intro to the course + Prob. review
2	PGM foundations
3	PGM foundations II
4	Freq. vs Bayesian + Prob. Prog. + Mixture models
5	Regression models
6	Classification and Hierarchical models
7	Temporal models
8	Topic Models
9	Markov-chain Monte Carlo (MCMC)
10	Variational inference
11	Generative models
12	Gaussian processes
13	Project support

Model-based Machine Learning



Learning objectives

At the end of this lecture, you should be able to:

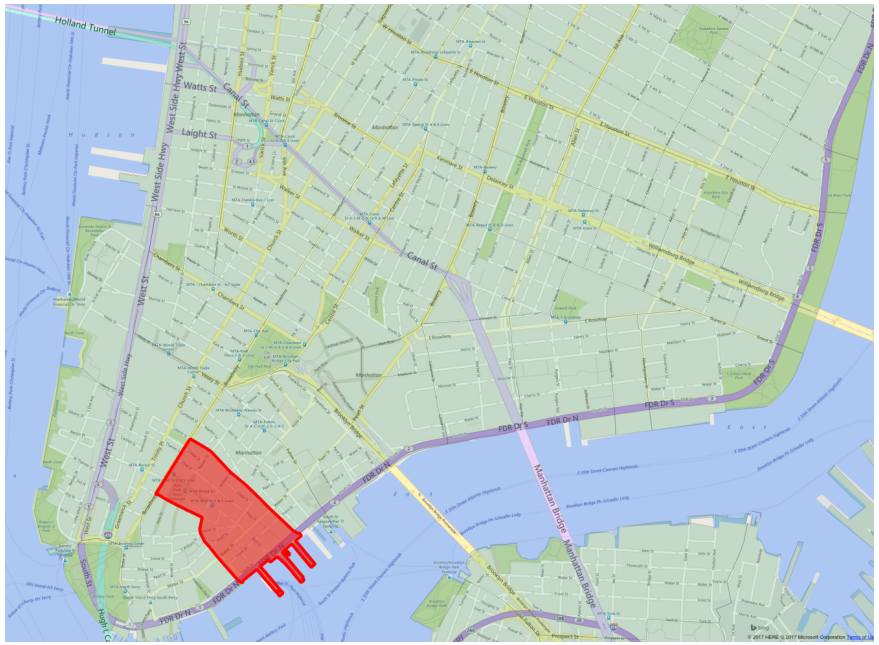
- Explain what (Bayesian) linear regression is and its underlying assumptions
- Modify the standard linear regression assumptions (e.g. Poisson likelihood or other) according to the data available
- Explain what an heteroscedastic model is, and how to model data-dependent observation noise
- Explain the basic ideas behind neural networks and what they allow you to do
- Relate different ways of modelling the dependency of a continuous random variable on other variables and justify their suitability for a problem, e.g.:
  - Linear-Gaussian relationship
  - Poisson with conditional rate
  - Gaussian with data-dependent noise
  - Combinations of linear models with (Bayesian) neural networks
- Implement the modelling techniques above in Pyro

Modeling taxi demand in New York City

- (Almost) all taxi trips in NYC from 2009 to mid 2016
- Original files have one trip per line
  - Pick-up location and time
  - Drop-off location and time
  - Other variables such as trip price and number of passengers
- Weather data from the National Oceanic and Atmospheric Administration
- Research question: **model taxi pickups across the city**
- Useful to optimize taxi service
  - Similar to many other demand problems (shared modes, public transport, energy, water, goods, communication...)

Modeling taxi demand in New York City (cont'd)

- Preprocessed data
  - Grouped data by census tract in 1 hour intervals
  - Extended grouped taxi data with relevant weather information
- Case study: **Wall Street**



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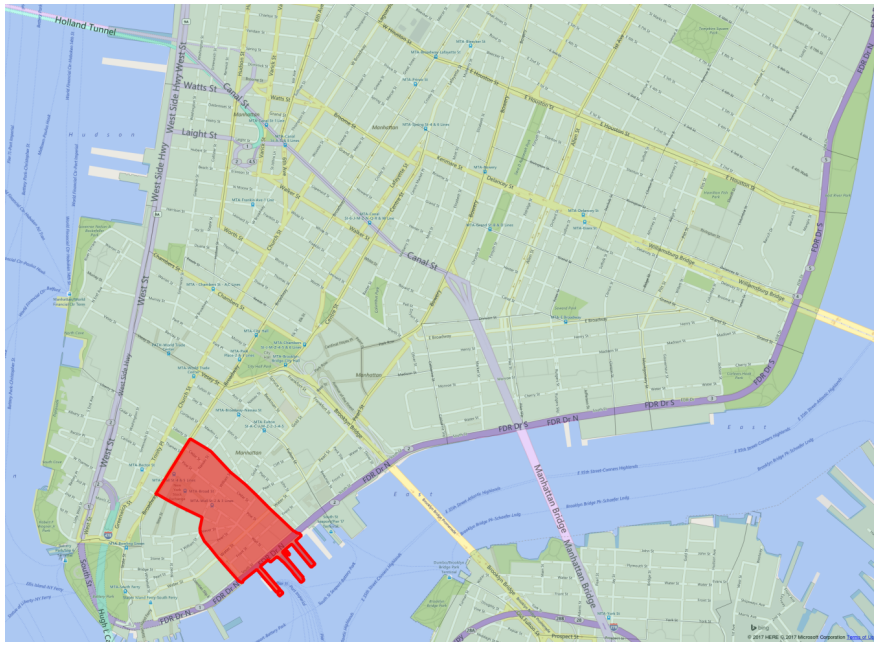
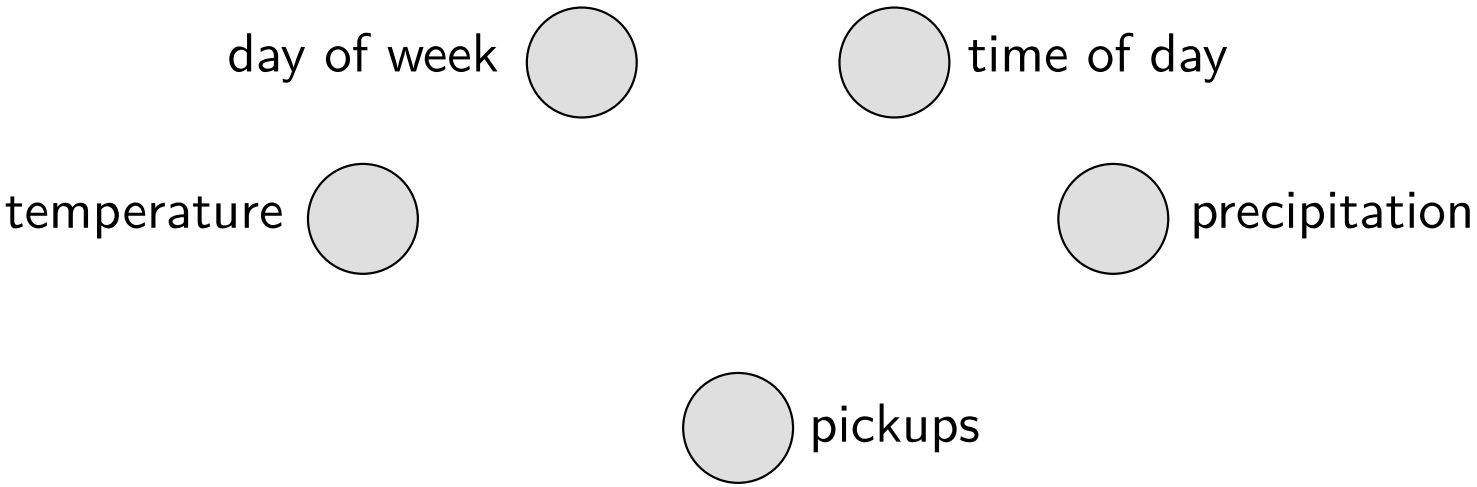


- What we know: day of the week, time of the day, temperature, precipitation, etc.
- Target variable: number of taxi pickups

Modeling taxi demand in New York City (cont'd)

- Let's start thinking about the graphical model...



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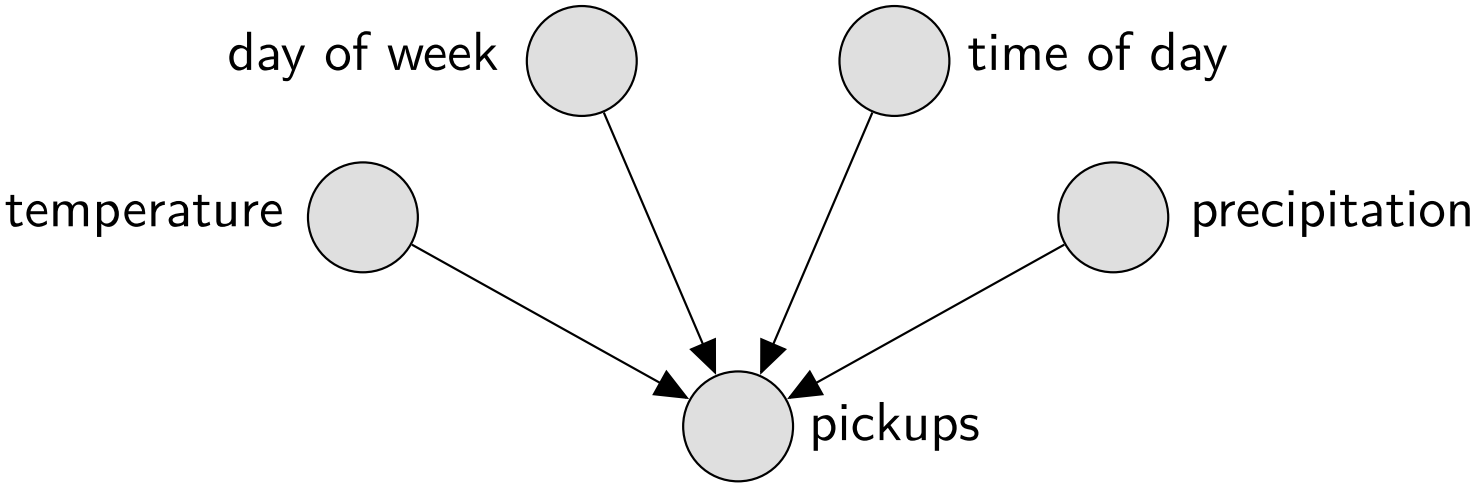

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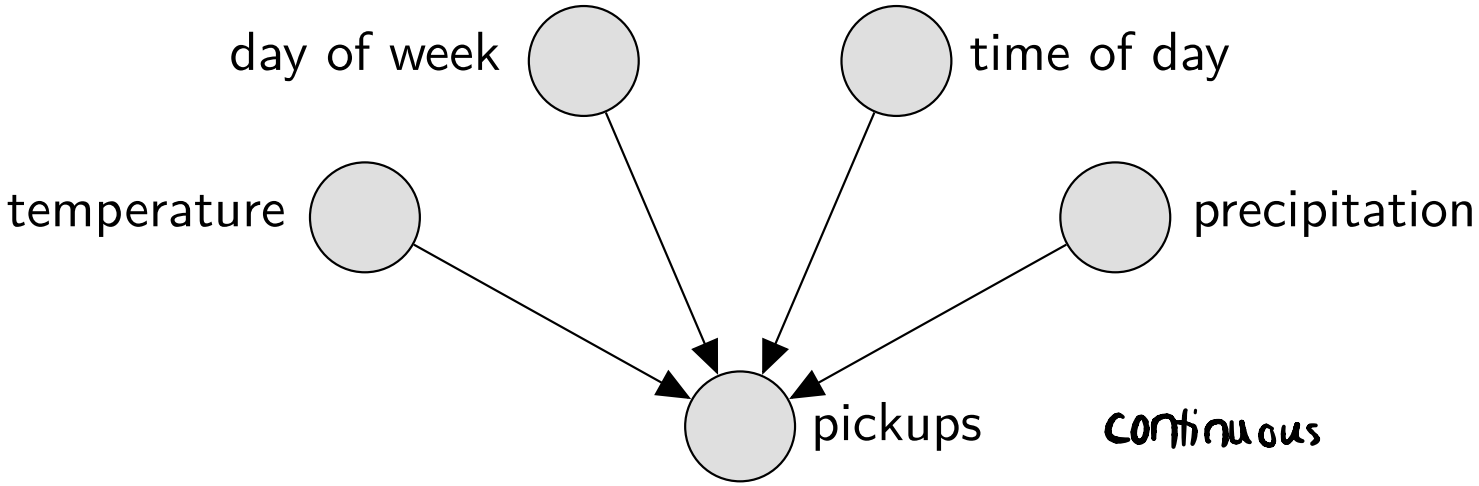

- What distribution should we assign to the pickups variable?
- How should we model the dependency of the pickups on the other variables?
- Do we need to assign distributions to these other variables (i.e. temperature, day of week, time of day, etc.)?
- This puts us right into the **regression** framework!

Regression

- Regression - predict response variable y from a collection of D predictor variables $x_1, x_2, \ldots, x_d, \ldots, x_D$

  y - target, response or dependent variable
  x_d - feature(s), covariate(s), explanatory or independent variable(s)

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- A few examples:
  - travel time prediction
  - predicting demand for autonomous vehicles
  - temperature/rainfall forecast
  - estimation of audience to a concert
  - prediction of future values of a share or a commodity (e.g. petrol)
  - prediction of house prices, number of voters in a state, births in a year
  - and, of course, predicting taxi demand!

Linear regression

- The dependent variable y is a function of all the predictor variables

$$y = f(x_1, x_2, \ldots, x_d, \ldots, x_D)$$

- Ok, but what function?
- Simplest approach is to assume a **linear relationship**

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_D x_D = \beta_0 + \sum_{d=1}^{D} \beta_d x_d$$

β_0 is the *intercept* (or bias) and $\{\beta_1, \beta_2, \ldots, \beta_D\}$ are the *coefficients* (or weights)

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- We can write this more compactly using **vector notation**

$$y = \beta_0 + \boldsymbol{\beta}^\mathsf{T} \mathbf{x}$$

  where $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_D)^\mathsf{T}$ and $\mathbf{x} = (x_1, x_2, \ldots, x_D)^\mathsf{T}$

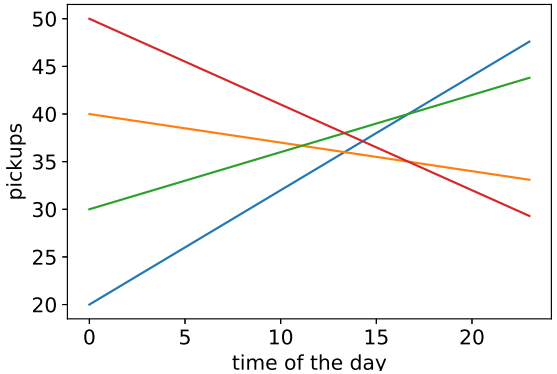
Note

The intercept β_0 can be seen as a coefficient for a special covariate x_0 that is always equal to 1. Thus, it is sometimes omitted.

Linear regression

- Linear assumption can seem naive...

$$y = f(\mathbf{x}) = \beta^{\mathsf{T}}\mathbf{x}$$



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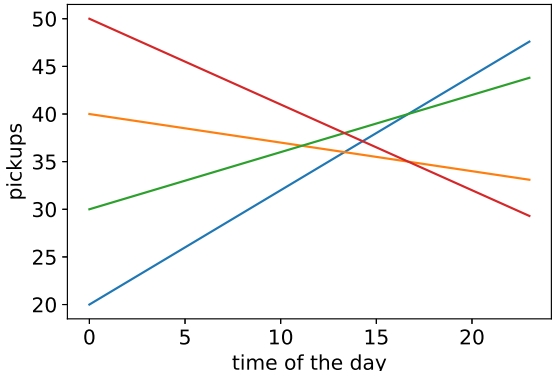

- But, the features $\mathbf{x}$ can be extremely **flexible**!
  - Any characteristic of the data
  - Indicator functions and 1-of-K encodings (e.g. $x_1 = \mathbb{I}[\text{weekend} = \text{True}]$)
  - Transformations of the original features (e.g. $x_2 = \log x_1$)
  - Basis expansion (e.g. $x_2 = x_1^2$ and $x_3 = x_1^3$) - polynomial fitting!
  - Interactions between features (e.g. $x_3 = x_1 x_2$)

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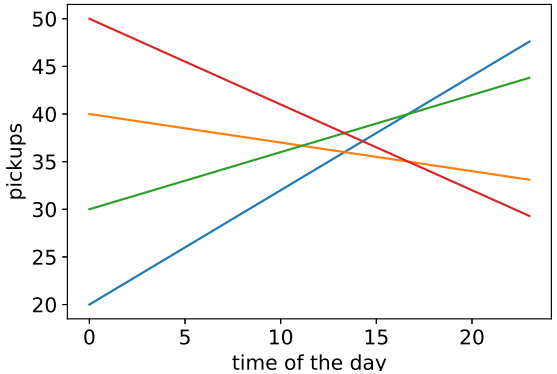

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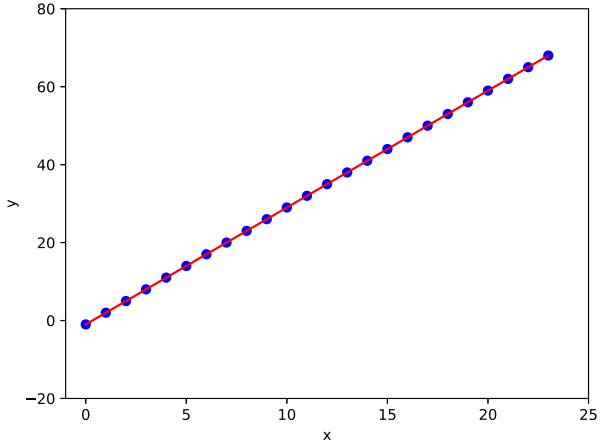
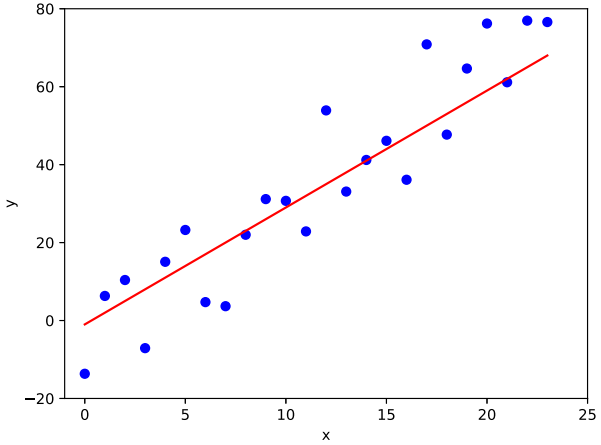
- Key aspects of linear regression
  - Simplicity
  - Flexibility
- One of the most important and widely used methods in statistics and machine learning!

Linear regression

- In practi



(a) dream world



(b) tough reality

- Add error term ϵ to account for observation noise

$$y = \beta^\mathsf{T}\mathbf{x} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

gaussian distribution

- We can equivalently write

$$\mathcal{N}(y|\underbrace{\beta^\mathsf{T}\mathbf{x}}_{\text{centered at } \beta^T x}, \sigma^2)$$

Linear regression as a graphical model

- We have a dataset $\mathcal{D}$ consisting of N observations of the targets y_n which depend on their corresponding explanatory variables $\mathbf{x}_n$

$$\mathcal{D} = \{\mathbf{x}_n, y_n\}_{n=1}^{N}$$

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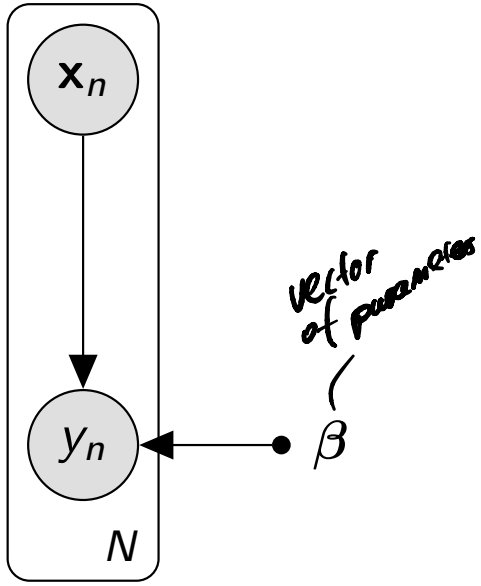

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  1. For $n = 1 \ldots N$ do:
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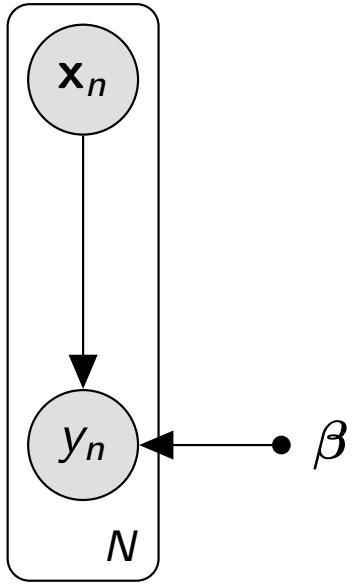

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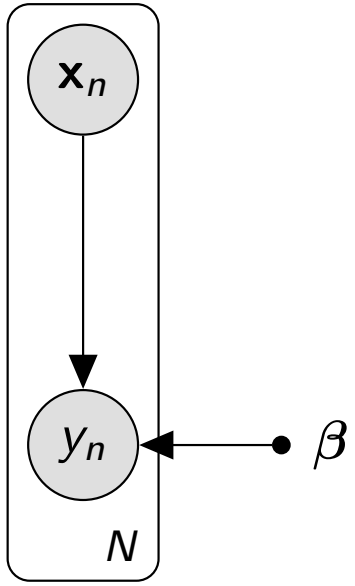

- Joint probability distribution factorizes as

$$p(\mathbf{y}|\mathbf{X}, \beta, \sigma) = \underbrace{\prod_{n=1}^{N} \mathcal{N}(y_n | \beta^\mathsf{T} \mathbf{x}_n, \sigma^2)}_{\text{likelihood}}$$

where $\mathbf{y} = \{y_n\}_{n=1}^N$, $\mathbf{X} = \{\mathbf{x}_n\}_{n=1}^N$, β are the model parameters and σ is fixed.

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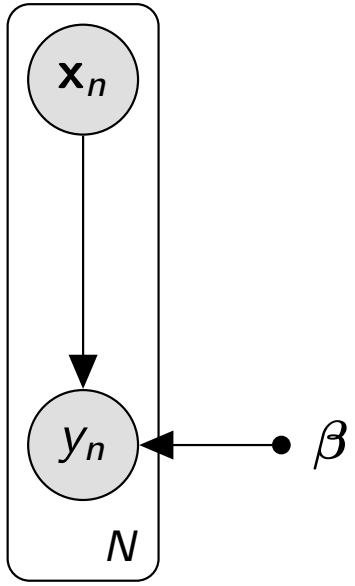

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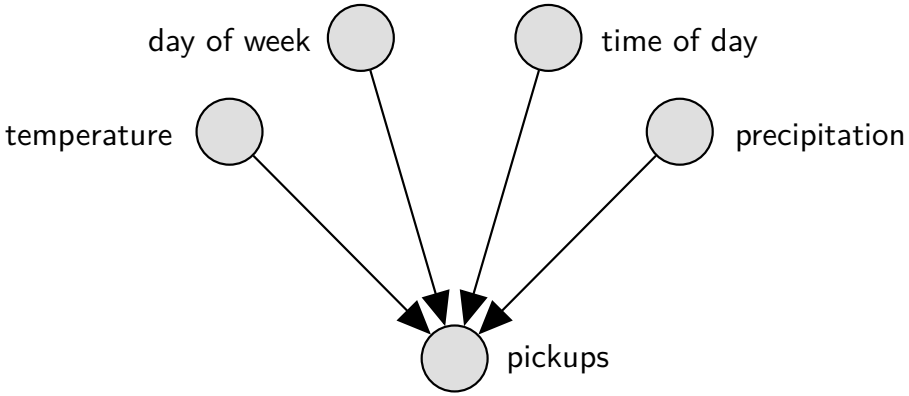
Note

We don't care about modeling $p(\mathbf{X})$. This is called a **conditional model** and contrasts with fully generative models.

Going back to our taxi demand case study...

- Let's revise the **modeling assumptions** that we made



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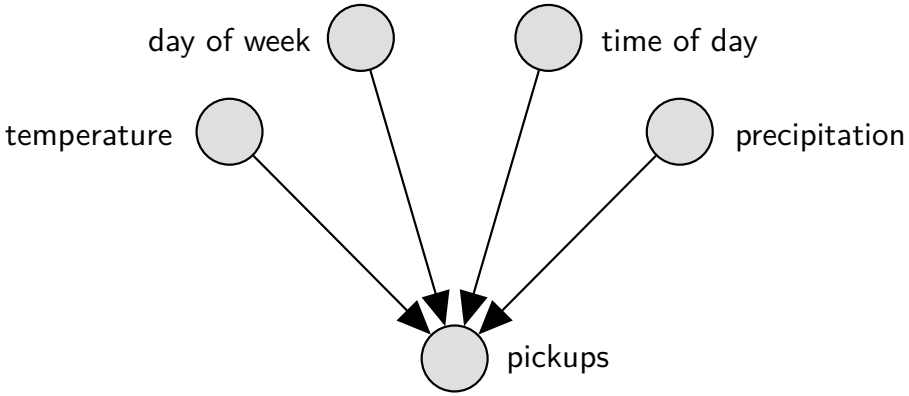

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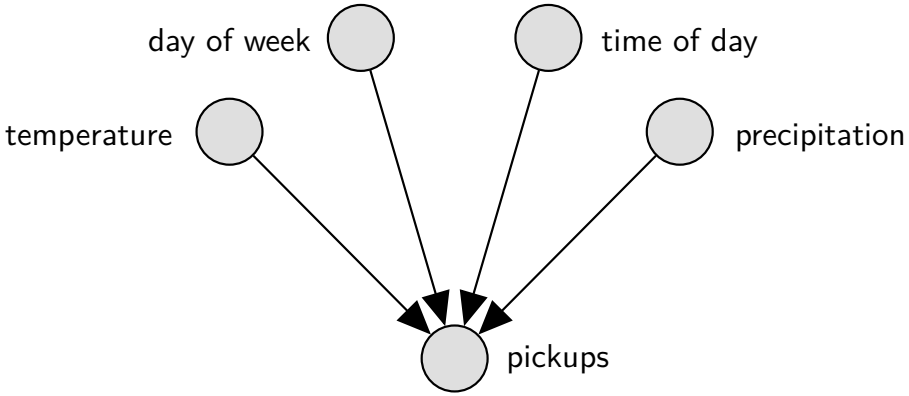

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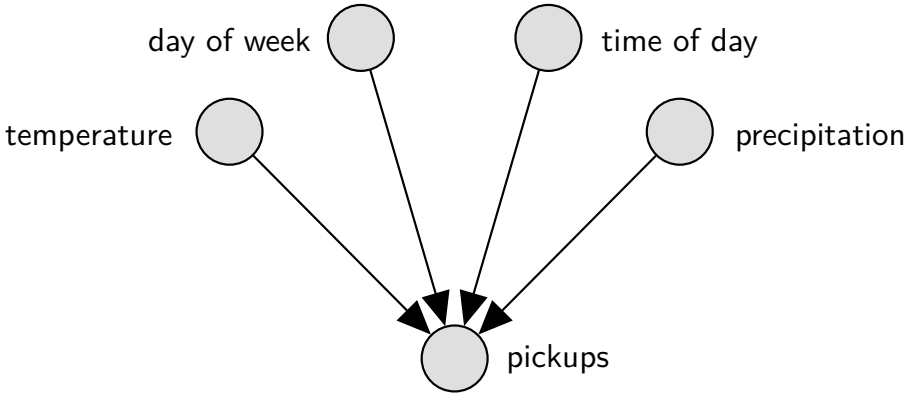

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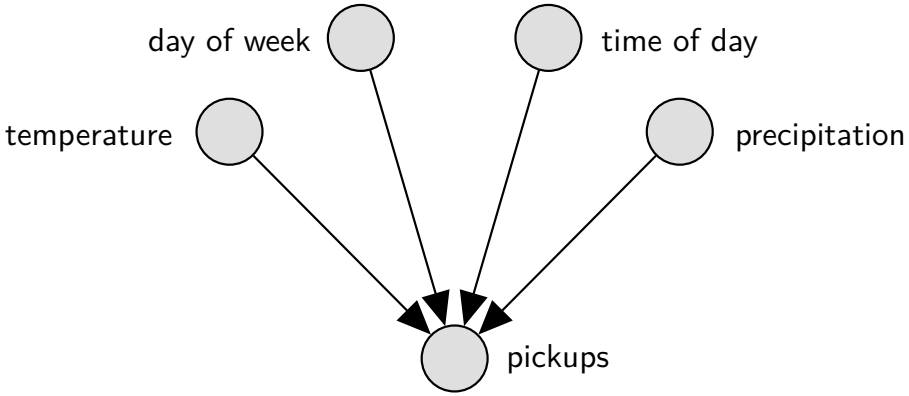

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  - In this case, no. They are always observed and we are only interested in modeling the behavior of the pickups variable

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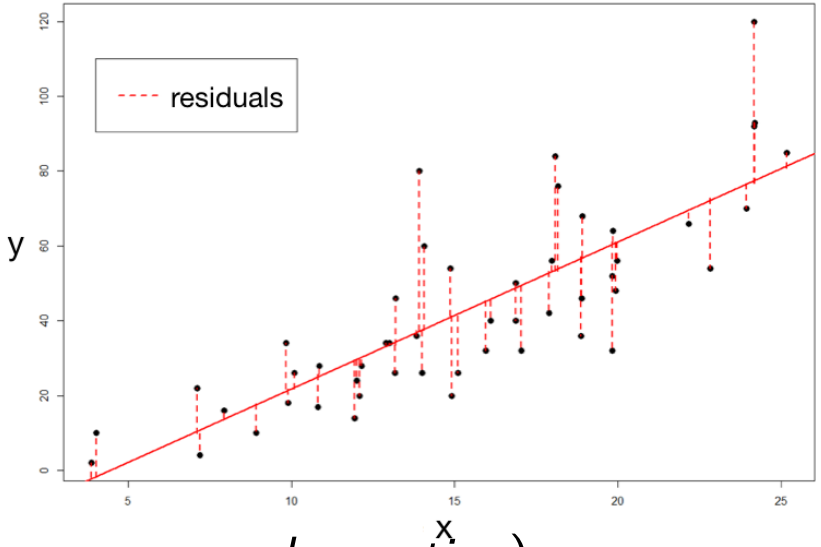
$$\hat{\beta} = \arg\min_{\beta} \sum_{n=1}^{N} \left(y_n - \beta^\mathsf{T}\mathbf{x}_n\right)^2$$



- Has a nice analytical solution (the famous *normal equation*)

$$\hat{\beta} = \left(\mathbf{X}^\mathsf{T}\mathbf{X}\right)^{-1}\mathbf{X}^\mathsf{T}\mathbf{y}$$

Model estimation: an alternative view

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Don't believe it?

Replace $\mathcal{N}(y_n | \beta^{\mathsf{T}} \mathbf{x}_n, \sigma^2)$ in the expression above by the definition of the Gaussian, take the derivative w.r.t. β, set it to zero and solve for β.

- This is called **maximum likelihood estimation (MLE)**!
- It allows to find a **point estimate** for the parameters in a probabilistic model

Adding priors

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  - What if we have some prior knowledge on the values of β?
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- A typical choice is a Gaussian with zero mean and a diagonal covariance matrix with λ in the diagonal elements

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- This encourages the values of β to be centered around zero with more or less variance depending on λ. But be careful:
  - Too large λ may lead **overfitting** (neutralizes the benefit of the prior)
  - Too small λ may lead **underfitting** (constrains the model too much)

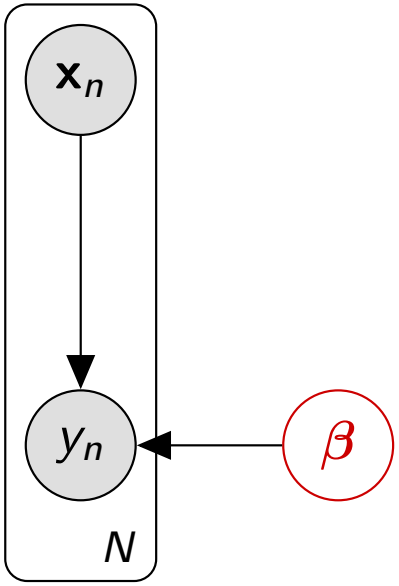
Bayesian linear regression model

- Updated graphical model

1) Draw $\beta \sim \mathcal{N}(\beta \mid \vec{0}, \lambda \vec{I}$

2) For each feature vector $\vec{x}_n$

Draw target $y_n \sim \mathcal{N}(y_n \mid \beta^T \vec{x}_n, \sigma^2)$



Bayesian linear regression model

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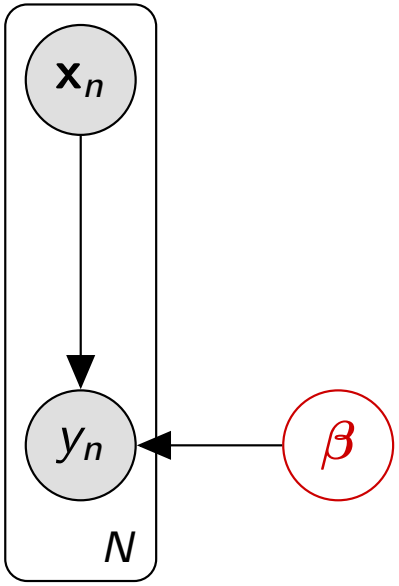

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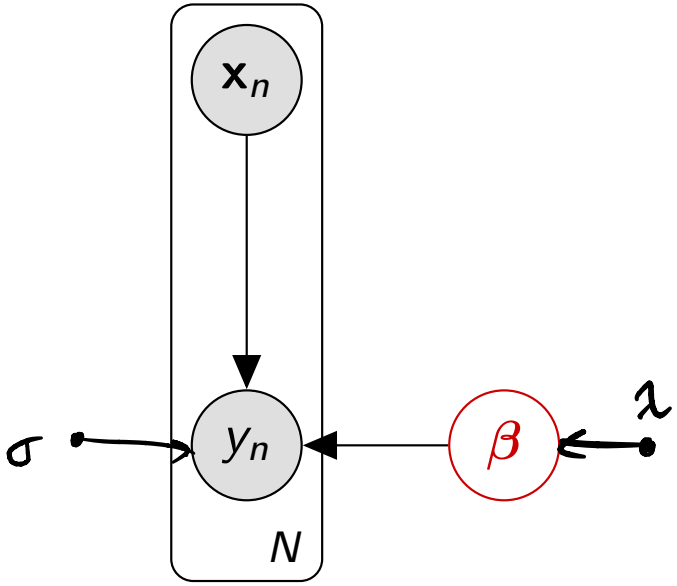

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- Joint probability distribution now factorizes as

given

$$p(\mathbf{y}, \beta|\mathbf{X}, \underbrace{\sigma, \lambda}_{\text{hyperparameters}}) = p(\beta|\lambda) \prod_{n=1}^{N} p(y_n|\beta, \mathbf{x}_n, \sigma)$$

$$= \underbrace{\mathcal{N}(\beta|\mathbf{0}, \lambda\mathbf{I})}_{\text{prior}} \times \underbrace{\prod_{n=1}^{N} \mathcal{N}(y_n|\beta^\mathsf{T}\mathbf{x}_n, \sigma^2)}_{\text{likelihood}}$$

Inference

- **Goal:** compute **posterior** distribution on β
- Following Bayes' theorem

Gaussian x Gaussian

$$\underbrace{p(\beta|\mathbf{y}, \mathbf{X}, \sigma, \lambda)}_{\text{posterior}} \propto \underbrace{\mathcal{N}(\beta|\mathbf{0}, \lambda\mathbf{I})}_{\text{prior}} \times \underbrace{\prod_{n=1}^{N} \mathcal{N}(y_n|\beta^\mathsf{T}\mathbf{x}_n, \sigma^2)}_{\text{likelihood}}$$

- We can find an analytical solution to this - exact inference is possible!
- We will cover inference methods later in the course...
- For now, Pyro will take care of it for us :-)

Model estimation: MAP

- Alternatively to computing the posterior distribution on β, we can find a point estimate by maximizing the (log) joint probability of the model

$$\hat{\beta} = \arg\max_{\beta} \log \left(\mathcal{N}(\beta|\mathbf{0}, \lambda\mathbf{I}) \prod_{n=1}^{N} \mathcal{N}(y_n|\beta^{\mathsf{T}}\mathbf{x}_n, \sigma^2) \right)$$

$$= \arg\max_{\beta} \left(\log \mathcal{N}(\beta|\mathbf{0}, \lambda\mathbf{I}) + \sum_{n=1}^{N} \log \mathcal{N}(y_n|\beta^{\mathsf{T}}\mathbf{x}_n, \sigma^2) \right)$$

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$$= \arg\max_{\beta} \left(\underbrace{\log \mathcal{N}(\beta|\mathbf{0}, \lambda\mathbf{I})}_{\text{replace with pdf}} + \sum_{n=1}^{N} \log \mathcal{N}(y_n|\beta^{\mathsf{T}}\mathbf{x}_n, \sigma^2) \right)$$

- This is called **maximum-a-posteriori (MAP)** estimation

Note

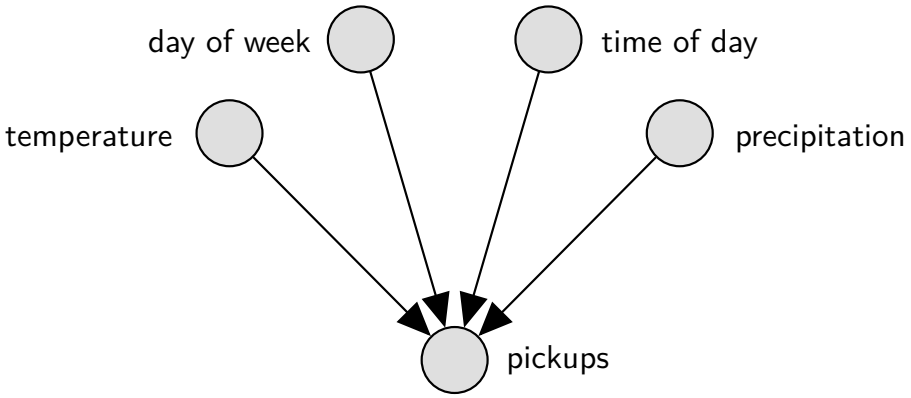
This is just like the MLE estimator plus a new term: $\log \mathcal{N}(\beta|\mathbf{0}, \lambda\mathbf{I})$. It penalizes the coefficients β for getting too large (overfitting). This is called **regularization**. See the derivation in Appendix.

Playtime!

- Ancestral sampling from linear regression model
  - See “04 - Regression models - Part 1.ipynb" notebook
  - Expected duration: 15 minutes
- Linear regression model of taxi pickups in NYC
  - See “04 - Regression models - Part 2.ipynb" notebook
  - Do section 2.1
  - Expected duration: 1 hour

Going back to our taxi demand case study...

- Let's revise (again) the **modeling assumptions** that we made



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- But is this really the most appropriate distribution in this case?

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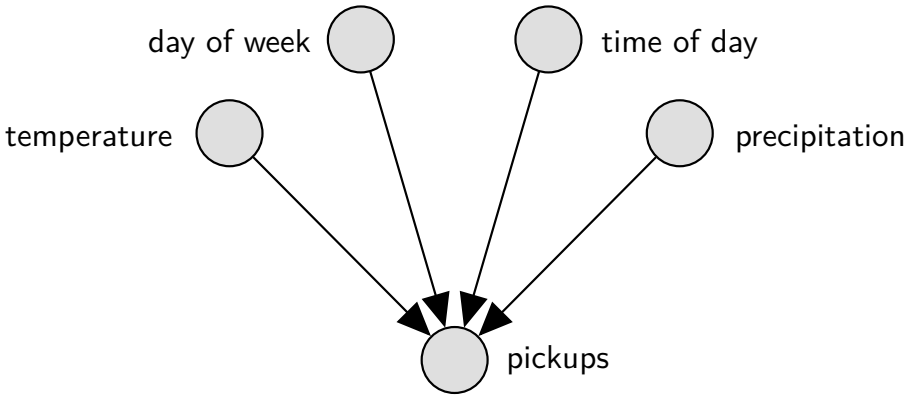

- What distribution should we assign to the pickups variable?
  - Gaussian
- But is this really the most appropriate distribution in this case?
  - Number of pickups is a count: $y_n \in \mathbb{N}$
  - A common distribution for modelling count data is the **Poisson**

Poisson regression

- “Poisson distribution expresses the probability of a given number of events occurring in a fixed interval of time and/or space if these events occur with a **known constant rate** and independently of the time since the last event"

Poisson regression

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- Sounds appropriate to model taxi pickups, but the rate is both **unknown** and **non-constant**...
  - As we did for the Gaussian, we can make the rate of the Poisson linearly dependent on the features $\mathbf{x}$

$$y_n \sim \text{Poisson}(y_n | \beta^\top \mathbf{x}_n)$$

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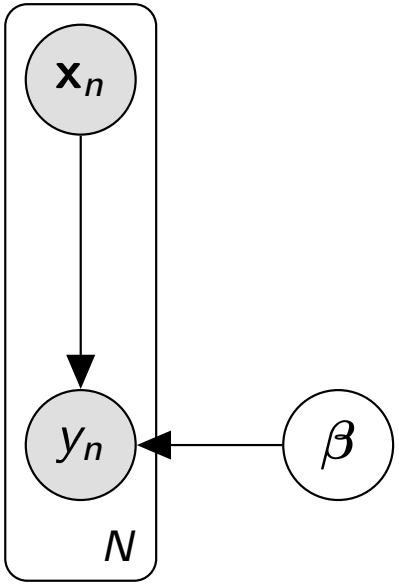
Note

Link functions get their names from the inverse of the transformation.

Bayesian poisson regression model

- Graphical model looks the same as before



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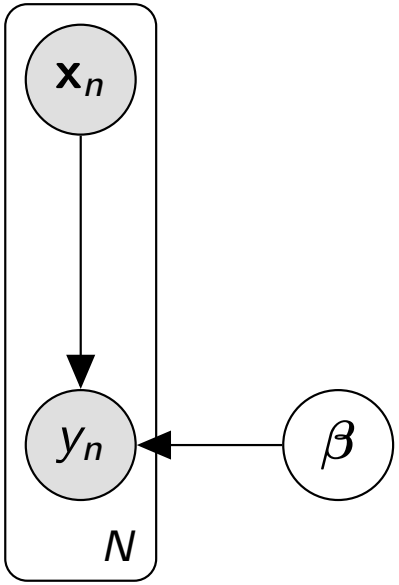

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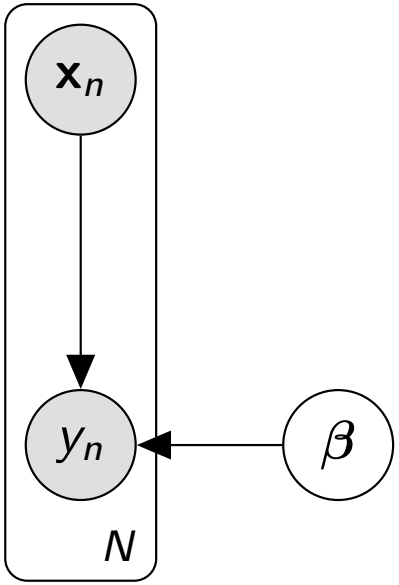

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- Joint probability distribution becomes

$$p(\mathbf{y}, \beta|\mathbf{X}, \lambda) = p(\beta|\lambda) \prod_{n=1}^{N} p(y_n|\beta, \mathbf{x}_n)$$

$$= \underbrace{\mathcal{N}(\beta|\mathbf{0}, \lambda\mathbf{I})}_{\text{prior}} \times \underbrace{\prod_{n=1}^{N} \text{Poisson}(y_n|e^{\beta^\mathsf{T}\mathbf{x}_n})}_{\text{likelihood}}$$

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- Exact inference is no longer tractable
- Must resort to **approximate inference** methods
- Not a problem for Pyro :-)

Playtime!

- Poisson regression model of taxi pickups in NYC
- See “04 - Regression models - Part 2.ipynb" notebook
- Do part 2.2
- Expected duration: 30 minutes

Going back to the modelling assumptions...

- Suppose that Gaussian was indeed the most appropriate distribution for the target variable y

$$y_n \sim \mathcal{N}(y_n | \beta^\mathsf{T} \mathbf{x}_n, \sigma^2)$$

- What if our observations of y had **non-constant noise**?

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- Suppose that Gaussian was indeed the most appropriate distribution for the target variable y

$$y_n \sim \mathcal{N}(y_n | \beta^\mathsf{T} \mathbf{x}_n, \sigma^2)$$

- What if our observations of y had **non-constant noise**?
- Consider the problem of modelling traffic speed data from probe vehicles
- The assumption of constant observation noise σ^2 might be too strong!

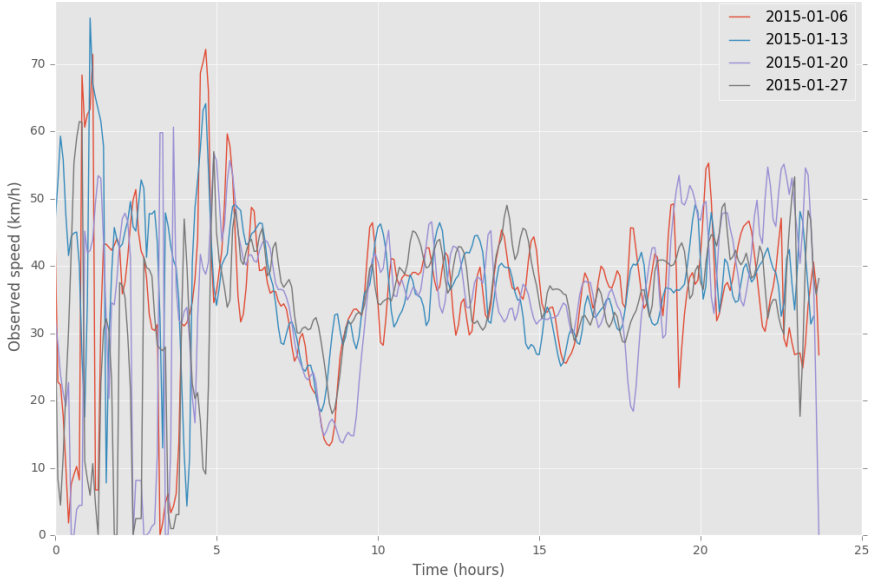

Figure: Traffic speeds in a road segment in Nørreport

Heteroscedastic regression

- We can relax the constant observation noise assumption by making the variance (linearly) dependent on a set of arbitrary features $\mathbf{u}$ (e.g. time of the day)

$$y_n \sim \mathcal{N}(y_n | \boldsymbol{\beta}^\top \mathbf{x}_n, e^{\boldsymbol{\eta}^\top \mathbf{u}_n})$$

where $\boldsymbol{\eta}$ is a new set of coefficients to parameterize the relation between the features $\mathbf{u}$ and the observation noise

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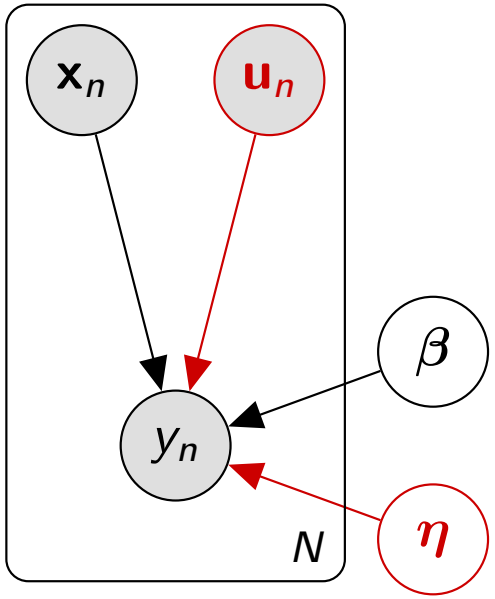
- As with the Poisson, we use a log link function to ensure non-negative variances

$$e^{\boldsymbol{\eta}^\top \mathbf{u}_n} \in (0, \infty)$$

- This allows to account for things like **time-varying observation noise** and produce better **uncertainty estimates** for the predictions!
  - Useful to know how reliable the predictions are

Bayesian heteroscedastic regression model

- Updated graphical model



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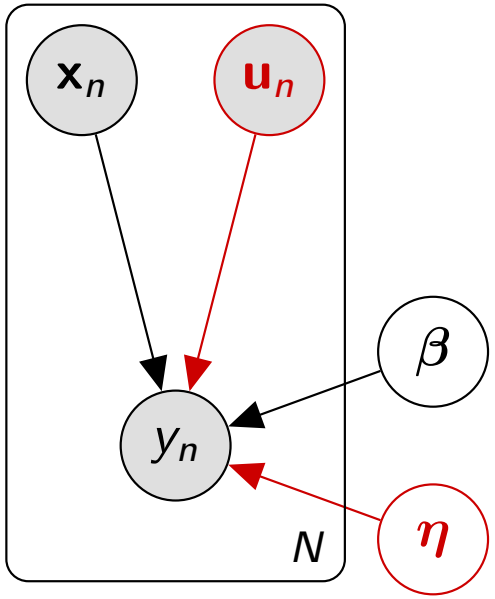

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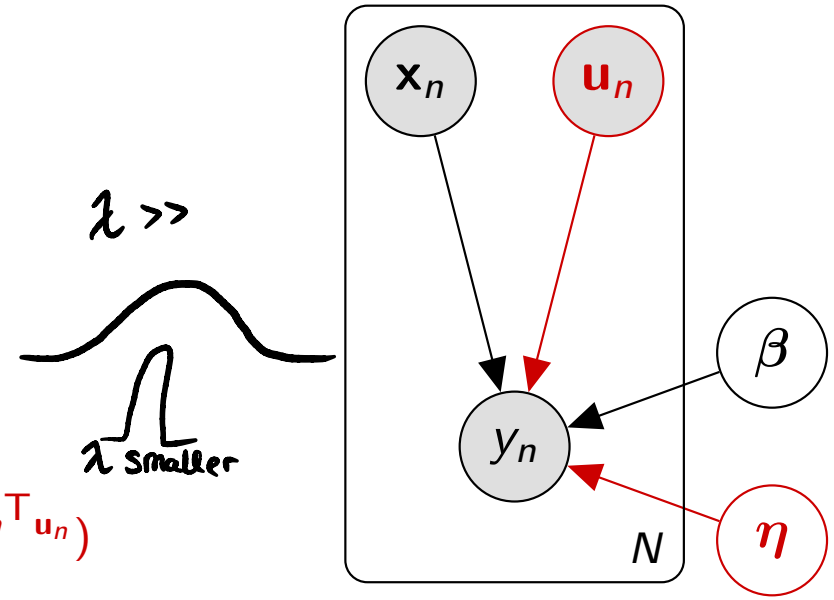

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Playtime!

- Heteroscedastic model of taxi pickups in NYC
- See “04 - Regression models - Part 2.ipynb" notebook
- Do part 2.3

Beyond linearity...

- So far we have been assuming a linear relationship between y_n and $\mathbf{x}_n$, such that

$$y_n = f(\mathbf{x}_n) + \epsilon$$

  where $f(\mathbf{x}_n) = \beta^\mathsf{T}\mathbf{x}_n$
- As previously explained this is far more powerful than it looks at first sight!
- However, in some cases, it might still not be enough... But what can we do?

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- As previously explained this is far more powerful than it looks at first sight!
- However, in some cases, it might still not be enough... But what can we do?
- **Gaussian processes** (GPs) allow us to model **non-linear** relationships!
  - **Non-parametric** models
  - Provide a probability distribution over functions
  - Place Gaussian process prior on the function f: $f \sim \mathcal{GP}$
  - GP prior specifies characteristics of the function, like stationarity, smoothness, periodicity, etc.
  - We will talk about GPs later on in the course! :-)
  - "Gaussian Processes for Machine Learning" book is a great resource[1]
  - Also, check out the Pyro's manual if you're interested

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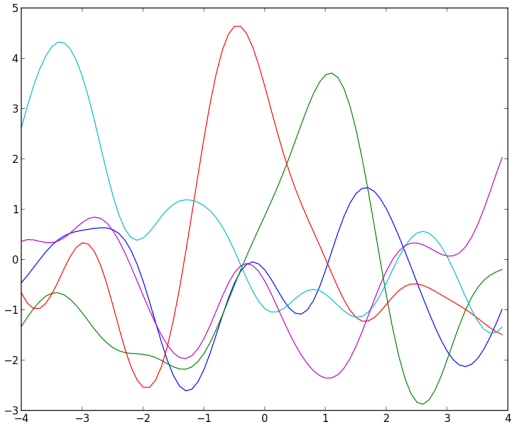
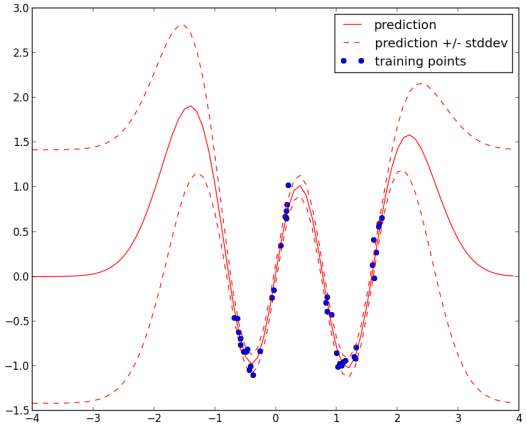
Beyond linearity...

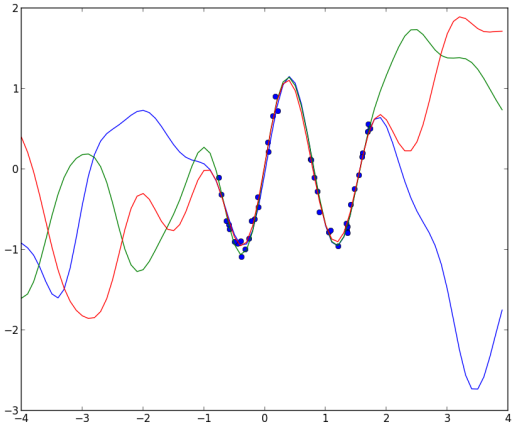
NNet



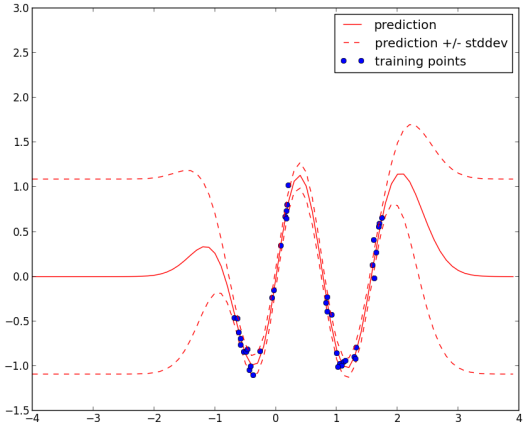
(a) samples from the GP prior



(b) predictive posterior



(c) samples from the GP posterior



(d) pred. post. after hyper-param. optimization

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[2] Paper: `https://arxiv.org/abs/1411.2581`

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- We can assume $f(\mathbf{x}_n)$ to be a complex **neural network (NNet)**!
  - In fact, we can parametrize any exponential family distribution using a NNet - check out, e.g.: *Deep Exponential Families*[2]
  - Intersection between Deep Learning and Bayesian methods is currently a very popular research topic!

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  - Intersection between Deep Learning and Bayesian methods is currently a very popular research topic!
- We can **combine PGMs with NNets**!
  - Variables in a PGM can be (part of) the input to a NNet
  - Output variables of a NNet can be part of a PGM

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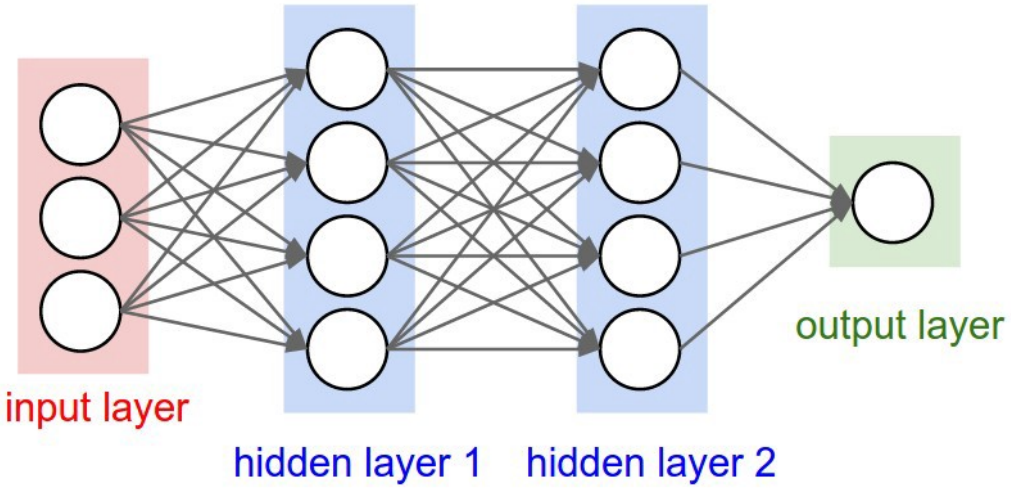
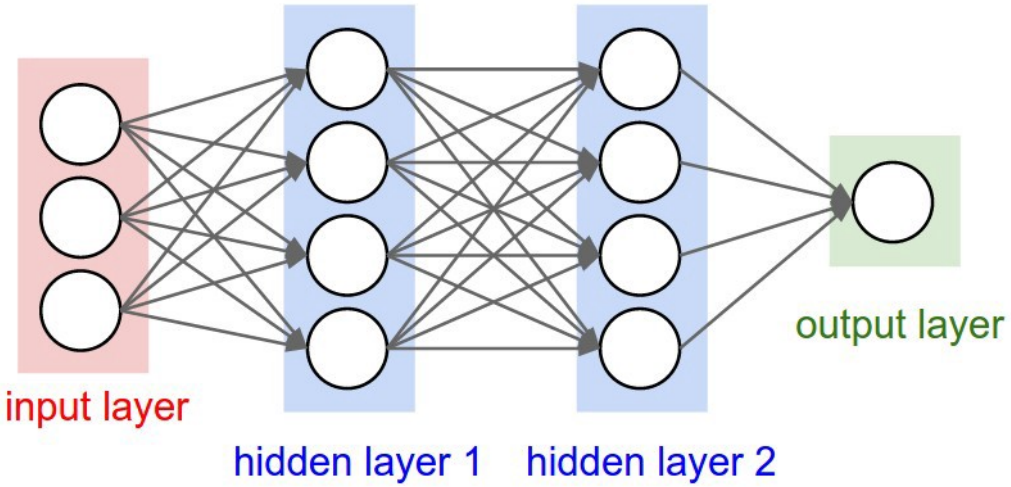
Crash course on (fully-connected/dense) Neural Networks



- Looks familiar doesn't it? :-)
  You can think of a neural networks as a PGM with some special characteristics!
- Each node is called a **neuron** and it computes a non-linear function of its inputs
- There is a weight w associated with each connection between neurons
- We **stack multiple layers** to obtain increasingly complex functions of the inputs
- More complex architectures exist, but are out of the scope of this course[3]

[3] Check "Deep Learning" course: `http://kurser.dtu.dk/course/02456`

Crash course on (fully-connected/dense) Neural Networks



- A neuron $h_i^{(l)}$ in layer l computes a weighted sum of its inputs from the previous layer $l-1$, and passes the result through a non-linearity (e.g. sigmoid, tanh,...)

$$h_i^{(l)} = \tanh\left(b_i^{(l)} + \sum_j w_{i,j}\, h_j^{(l-1)} \right)$$

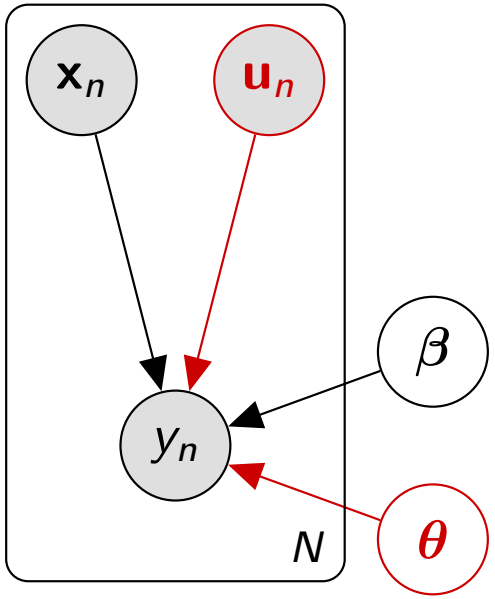
  where $b_i^{(l)}$ is a bias parameter and $w_{i,j}$ is the weight of the connection between $h_i^{(l)}$ and $h_j^{(l-1)}$
- More compactly using vector notation: $\mathbf{h}^{(l)} = \tanh(\mathbf{b}^{(l)} + \mathbf{W}^{(l)}\, \mathbf{h}^{(l-1)})$

Combining PGMs with neural networks

- A very simple practical example:

$$y_n = f_{\text{linear}}(\mathbf{x}_n) + f_{\text{nnet}}(\mathbf{u}_n) + \epsilon$$

where $f_{\text{linear}}(\mathbf{x}_n) = \beta^\top \mathbf{x}_n$ and $f_{\text{nnet}}(\mathbf{u}_n)$ is a NNet with parameters denoted by $\boldsymbol{\theta}$ (i.e. $\boldsymbol{\theta}$ includes all bias parameters and weights of the NNet)



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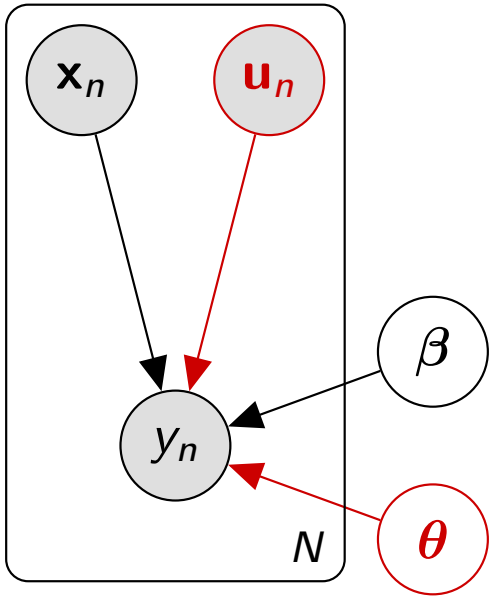
- Updated generative process
  1. Draw linear coefficients $\beta \sim \mathcal{N}(\beta|\mathbf{0}, \lambda\mathbf{I})$
  2. Draw NNet parameters $\boldsymbol{\theta} \sim \mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \tau\mathbf{I})$
  3. For the nth observation
     a. Draw target $y_n \sim \mathcal{N}(y_n|f_{\text{linear}}(\mathbf{x}_n) + f_{\text{nnet}}(\mathbf{u}_n), \sigma^2)$



Playtime!

- Combining PGMs with neural networks
- See “04 - Neural Network + Linear PGM - Pyro.ipynb" notebook

Appendix: Gaussian prior and regularization

derivation

- We have the following model (Bayesian linear regression with Gaussian prior):

$$p(\mathbf{y}, \beta | \mathbf{X}, \sigma, \lambda) = \mathcal{N}(\beta | \mathbf{0}, \lambda \mathbf{I}) \times \prod_{n=1}^{N} \mathcal{N}(y_n | \beta^\mathsf{T} \mathbf{x}_n, \sigma^2)$$

- We want to prove that the Gaussian prior component makes the MAP solution equivalent to the L2 regularization of the least squares solution. I.e.:

$$\arg\max_{\beta} \log \left(\mathcal{N}(\beta | \mathbf{0}, \lambda \mathbf{I}) \prod_{n=1}^{N} \mathcal{N}(y_n | \beta^\mathsf{T} \mathbf{x}_n, \sigma^2) \right)$$

$$= \arg\min_{\beta} \sum_{n=1}^{N} (y_n - \beta^\mathsf{T} \mathbf{x}_n)^2 + \gamma \beta^\mathsf{T} \beta$$

- For all $\lambda > 0$, and $\gamma = \frac{\sigma^2}{\lambda}$.

Appendix: Gaussian prior and regularization

derivation

- For simpler notation, let's assume $\hat{y}_n = \beta^\mathsf{T}\mathbf{x}_n$
- Let's start by calculating the log $p(\mathbf{y}, \beta|\mathbf{X}, \sigma, \lambda)$

$$\log\ p(\mathbf{y}, \beta|\mathbf{X}, \sigma, \lambda) = log\Big(\mathcal{N}(\beta|\mathbf{0}, \lambda\mathbf{I}) \times \prod_{n=1}^{N} \mathcal{N}(y_n|\beta^\mathsf{T}\mathbf{x}_n, \sigma^2)\Big)$$

$$= \log \frac{1}{(2\pi|\lambda\mathbf{I}|)^{\frac{1}{2}}} e^{\left(-\frac{1}{2}(\beta^\mathsf{T}(\lambda\mathbf{I})^{-1}\beta)\right)} + \sum_{n=1}^{N} \log \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}} e^{(-\frac{1}{2\sigma^2}(\hat{y}_n - y_n)^2)}$$

$$= \underbrace{\log \frac{1}{(2\pi|\lambda\mathbf{I}|)^{\frac{1}{2}}}}_{const.} - \frac{1}{2}(\beta^\mathsf{T}(\lambda\mathbf{I})^{-1}\beta) + \cancel{N \log 1} - \underbrace{\frac{N}{2}\log(2\pi\sigma^2)}_{const.} - \sum_{n=1}^{N} \frac{(\hat{y}_n - y_n)^2}{2\sigma^2}$$

$$= -\frac{1}{2}(\beta^\mathsf{T}(\lambda\mathbf{I})^{-1}\beta) - \sum_{n=1}^{N} \frac{(\hat{y}_n - y_n)^2}{2\sigma^2} + const.$$

Appendix: Gaussian prior and regularization

derivation

- So, we have:
  $\log\ p(\mathbf{y}, \beta|\mathbf{X}, \sigma, \lambda) = -\frac{1}{2}(\frac{\beta^\mathsf{T}\beta}{\lambda}) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(\hat{y}_n - y_n)^2 + const.$
- ...and we want to calculate $\arg\max_\beta \log\ p(\mathbf{y}, \beta|\mathbf{X}, \sigma, \lambda)$

$\arg\max_\beta \log\ p(\mathbf{y}, \beta|\mathbf{X}, \sigma, \lambda) = \arg\min_\beta -\log\ p(\mathbf{y}, \beta|\mathbf{X}, \sigma, \lambda)$

$= \arg\min_\beta -\log\ p(\mathbf{y}, \beta|\mathbf{X}, \sigma, \lambda) = \arg\min_\beta \frac{1}{2}(\frac{\beta^\mathsf{T}\beta}{\lambda}) + \frac{1}{2\sigma^2}\sum_{n=1}^{N}(\hat{y}_n - y_n)^2$
$= \arg\min_\beta \frac{1}{2\sigma^2}\left(\frac{\sigma^2}{\lambda}(\beta^\mathsf{T}\beta) + \sum_{n=1}^{N}(\hat{y}_n - y_n)^2\right)$

- If we equate $\gamma = \frac{\sigma^2}{\lambda}$, and rewrite $\beta^\mathsf{T}\mathbf{x}_n = \hat{y}_n$, we get the L2 regularized least squares formulation, as desired[4]:

$$\arg\min_\beta \sum_{n=1}^{N}(y_n - \beta^\mathsf{T}\mathbf{x}_n)^2 + \gamma\beta^\mathsf{T}\beta$$

Q.E.D.

[4] Note that this γ effectively corresponds to the usual λ parameter on the regularization literature, which is **not** the same as the λ in our prior, which is also common in literature!